

CAMP DAVID, MD, USA

WMO: 722823

Lat: 39.645N

Lon: 77.468W

Elev: 561

StdP: 94.76

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03749

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-12.3	-21.9	0.6	-13.7	-19.5	0.7	-10.5	12.0	-4.7	11.1	-3.4	5.2	330	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	30.4	20.8	29.0	20.3	27.7	20.0	23.3	28.0	22.4	26.8	21.8	25.7	2.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	17.6	25.6	21.1	16.9	24.6	20.5	16.2	23.9	71.8	28.0	68.6	26.8	66.0	25.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.5	7.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-1.9	-0.7	4.5	11.6	16.8	20.3	22.8	21.8	18.3	12.5	6.5	1.5
	DBStd	9.85	6.22	6.07	6.05	5.46	4.39	3.14	2.75	2.69	3.91	5.09	4.82	5.55
	HDD10.0	1333	372	302	191	42	3	0	0	0	0	31	124	268
	HDD18.3	3026	627	532	428	215	82	17	2	4	49	193	356	521
	CDD10.0	1782	3	3	21	88	213	309	396	367	249	108	18	6
	CDD18.3	434	0	0	0	12	34	76	139	113	48	11	0	0
	CDH23.3	2358	0	0	1	112	182	387	843	566	239	28	0	0
	CDH26.7	527	0	0	0	39	27	73	216	118	51	3	0	0
Wind	WSAvg	3.5	4.5	4.6	4.3	3.9	3.1	2.8	2.6	2.4	2.6	3.3	3.8	4.5
Precipitation	PrecAvg	1118	84	63	105	94	94	101	98	96	116	86	80	89
	PrecMax	1750	212	151	227	192	213	221	245	152	243	192	169	161
	PrecMin	726	31	7	12	43	22	32	44	38	0	12	13	16
	PrecStd	230	45	34	46	38	48	55	47	30	68	52	37	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	17.0	21.7	31.1	29.1	31.0	32.8	31.7	30.1	26.4	19.5	17.8
		MCWB	12.8	12.9	12.7	17.0	19.4	21.6	22.2	21.1	21.4	19.8	14.2	14.2
	2%	DB	12.9	13.6	18.7	26.5	27.1	28.5	30.5	29.4	28.2	23.5	17.0	15.3
		MCWB	9.9	9.4	11.5	15.3	18.4	20.6	21.7	19.8	20.7	18.0	11.7	12.7
	5%	DB	10.1	11.1	16.5	22.9	25.3	27.1	29.0	27.9	26.1	21.6	15.2	12.3
		MCWB	7.3	7.9	10.8	13.7	17.8	19.7	20.9	19.8	19.8	17.0	10.3	10.2
	10%	DB	6.9	8.0	14.0	20.0	23.5	25.6	27.5	26.5	24.0	19.9	13.5	9.3
		MCWB	4.6	4.5	9.6	12.6	16.7	19.0	20.7	19.7	19.1	16.3	9.5	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	13.8	14.8	18.1	22.1	23.5	24.4	24.3	23.2	21.4	15.4	15.5
		MCDB	14.6	15.6	18.3	27.1	27.6	28.9	29.7	29.1	28.3	23.8	18.2	16.8
	2%	WB	10.6	10.8	13.4	16.2	20.7	22.1	23.4	22.9	21.9	20.1	13.2	12.9
		MCDB	12.3	12.4	16.7	23.7	25.3	26.9	28.4	26.7	25.4	22.2	15.0	14.8
	5%	WB	7.6	7.7	12.0	15.1	19.7	21.4	22.7	21.9	21.0	18.6	11.8	10.5
		MCDB	9.2	10.9	15.8	21.5	23.4	25.5	27.2	25.2	23.9	21.0	14.0	11.9
	10%	WB	4.6	4.7	9.7	13.7	18.5	20.4	21.9	21.1	20.0	16.7	10.4	7.1
		MCDB	6.9	7.7	13.9	19.1	21.7	24.0	26.1	24.3	23.0	19.4	12.9	8.9
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.7	8.7	9.8	8.6	8.4	8.2	7.9	7.8	7.2	7.2	6.4
		MCDBR	9.0	10.2	11.3	12.5	10.2	9.6	9.3	9.2	9.4	8.3	8.6	9.1
		MCWBR	8.2	8.3	7.6	6.4	5.4	5.0	4.5	4.2	4.5	4.4	5.9	8.2
	5% WB	MCDBR	8.7	10.1	10.6	11.5	8.8	8.6	8.2	7.8	8.0	7.5	8.6	9.1
		MCWBR	8.5	9.1	8.0	6.9	5.0	4.9	4.3	4.3	4.1	4.4	7.2	8.5
Clear-Sky Solar Irradiance	taub	0.290	0.303	0.333	0.372	0.406	0.464	0.481	0.476	0.402	0.343	0.304	0.290	
	taud	2.465	2.407	2.406	2.319	2.248	2.115	2.087	2.082	2.268	2.444	2.518	2.498	
	Ebn at Noon	885	919	923	903	874	821	802	794	837	861	863	858	
	Edh at Noon	78	95	107	124	137	156	159	156	122	92	74	71	
All-Sky Solar Radiation	RadAvg	1.89	2.79	3.70	4.76	5.33	5.91	5.85	5.20	4.24	3.09	2.20	1.65	
	RadStd	0.16	0.33	0.23	0.39	0.58	0.46	0.34	0.27	0.55	0.35	0.20	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	+63

Nomenclature: See separate page